BostonDynamics

TEAM 3

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PROSPERITY IN ROBOTICS HINGES ON BRINGING **‘FEATURES PROVIDABLE’** AND **‘FEATURES DEMANDED’** TO THE SAME PAGE; NO MORE, NO LESS

Societal

- Public perception of humanoid robots remains **mixed and uncertain;** the consensus is still evolving as exposure to these technologies increases.
- The uncanny valley phenomenon continues to influence public reaction, causing discomfort as robots approach human likeness without achieving it fully.
- There is a prevalent concern that the rise of smart robots poses a **threat to human employment,** potentially leading to structural unemployment and a growing socioeconomic divide.
- As robotic adoption accelerates, there's growing concern about preserving human-centric experiences in daily life.

Technology

- Achieving true autonomy in humanoid robots faces major technical challenges, as **current system struggles with cohesive framework.**
- The feasibility of developing the required advancements in manufacturing processes, and logistics optimization to **support such large-scale production is still unclear.**
- The increasing connectivity and intelligence of robotic systems introduce significant systems cybersecurity and data privacy concerns.
- There is a huge risk that AI-powered weapons could exhibit bias and discriminate based on factors like race, gender, or disability due to flawed or biased training data.

Economic

- The shift from industrial to consumer robotics presents challenges of establishing new supply chains, distribution channels, and business models for a largely untested market.
- Competitive pricing from companies like Unitree could disrupt the industry’s cost structure.
- The adoption of advanced robotics is seen as a driver to revitalize productivity growth across various sectors of the economy.
- This growing cost-effectiveness compared to human labor is driving increased adoption of robotic systems in manufacturing, logistics, and other sectors.

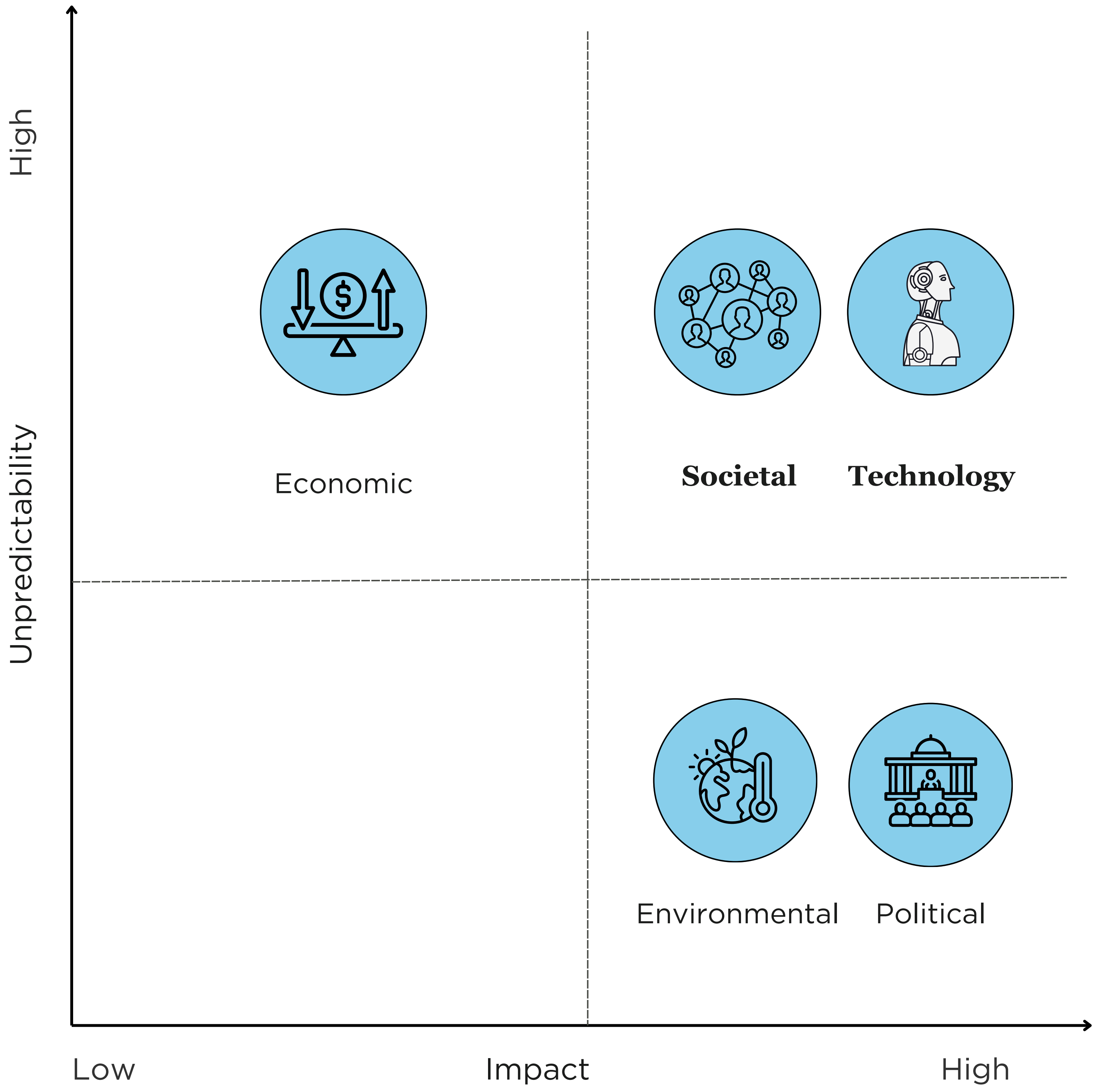
Environmental

- The rapid proliferation of robots and AI systems raise concerns about energy use, e-waste, and resource depletion.
- Sustainability is driving the adoption of Green Robotics, aligning technological advancement with ecological preservation.
- Autonomous robots, including drones and underwater vehicles, are increasingly used for environmental monitoring and conservation efforts.
- High-precision sensors and control algorithms enable robots to perform tasks with unparalleled accuracy, optimizing raw material usage and minimizing errors.

Political

- As advanced robotics begin to disrupt traditional industries and social structures, governments and policymakers struggle to formulate strategies.
- There is a global scramble to create unified standards and regulations.
- The European Commission's planned EU-wide robotics strategy for early 2025 reflects a cautious approach, aiming to ensure responsible and ethical deployment of robots.
- Landmark AI regulations aim to limit automated decision-making affecting employees, mandate human oversight, and require transparency in employment decisions.

BOSTON DYNAMICS NEEDS TO **HARBOR AND NOURISH PUBLIC PERCEPTION** AS THEY ASPIRE TO MAKE THEIR **TECHNOLOGICAL LEAPS**



Evolving Dynamics of Societal Acceptance

Overwhelming Acceptance

- Strong market interest but disruptors create resistance in adoption as competitors and overall industry witness dynamic shifts.

Acceptance Saturation

- Strong concerns about privacy, security, and ethical implications slowing adoption and consequently leading to stricter regulations.

Progress Toward Unified Robotic Integration

A Tech-Blitz

- A seamless fusion of hardware, software, and AI forms a unified operational model which will take us a step further into the Robotic Revolution.

Incremental Developments

- Existing fragmented systems leave space for some behavioral disparity (compared to humans) and pose challenges in scaling and implementing on a large-scale until dealt with.

IN AN AWAITING MARKET WITH THE CAPABILITY TO DELIVER, **A B2C MODEL** EMERGES AS A VIABLE ET LUCRATIVE OPTION

High Societal Acceptance

UNFULFILLED ROBOTIC PROMISES

People are not only asking for robots, they are begging for them. Initial promises and product testing has shown that people are ready but the technology simply does not arrive soon enough. Humanoid robots look like humans, act like them, but are not at the point of being an actual, useful companion. Companies are still pouring a lot of money into it, but it seemed they underestimated how hard this would be to achieve.

THE ROBOTIC REVOLUTION

Humanoid robots are part of the day-to-day life, helping with at-home tasks and mundane tasks. Robots are at a point were their decision making capabilities are at a autonomous level, and can perform like humans at an excellent level. They have integrated well to the current state of our society, and have been seen as a massive help for job creation and productivity. Companies are catering to all human needs, and focusing heavily on B2C.

Evolving Dynamics of Societal Acceptance

Slow
Incremental Integration

Progress Toward Unified Robotic Integration

Rapid
Unified AI and Hardware

ROBOTS IN MOVIES ARE GOOD ENOUGH

After money was put into the B2C market, it wasn't enough to create a solution that worked like the promises around humanoid robots. Humans are complex, and the hurdles behind providing a unified solution that isn't just for repetitive, mundane task, was not possible. Companies also saw how the general public reacted against the idea of humanoid robots, and the concerns across the public were very high. B2B seems like the best option.

HUMAN VS BOT

Technological advances were made and private companies developed humanoids that excel at their different use cases. However, humans were not ready to socially accept them to their lives just yet, and polarizing views emerged. There is social discomfort even when there are clear benefits, as it is the closest technology has been of displacing an actual human. Although the technology is top of the line, our World is still designed for and by humans.

Low Societal Acceptance

A ROBOTICS COMPETITOR'S SUCCESS WILL DEPEND ON QUICKLY ALIGNING **TECH CAPABILITIES, PEOPLE'S NEED AND PRICEPOINTS**, ALL IN ONE FRAME

High Societal Acceptance

Low Societal Acceptance

Slow Incremental Integration ← Progress Toward Unified Robotic Integration → Rapid Unified AI and Hardware

Evolving Dynamics of Societal Acceptance

TECH LEADERS MISSING THE USER CONNECTION

Insurgents:

There is a need and a demand for the advanced robots but these new companies fail to deliver a game changing product

MWbotics, Agility Robotics, APPTRONIK

Incumbents:

They struggle to advance in their own niches let alone build tech for advanced robots

ABB, KUKA, FANUC Robotics

Adjacents:

Their tech capabilities turns out to be limiting which does not meet the demands of the users

TESLA, NVIDIA

INNOVATORS BRIDGING TECHNOLOGY AND MARKET NEEDS

Insurgents:

Rapid innovation in these companies with funds put into invested by big players, starting with mass adoption in the B2B space

MWbotics, Agility Robotics, APPTRONIK

Incumbents:

They start building highly intelligent robots and penetrating the market they are already established

ABB, KUKA, FANUC Robotics

Adjacents:

Use their expertise in building deep tech and expand into the robots markets, which they have already showed signs of.

TESLA, NVIDIA

ESTABLISHED PLAYERS STUCK IN THEIR COMFORT ZONE

Insurgents:

Do not find much success in building the tech better, and not markets where customers would adopt these

MWbotics, Agility Robotics, APPTRONIK

Incumbents:

They focus on their core business rather than pushing the limit of the tech, specifically in humanoid robots

ABB, KUKA, FANUC Robotics

Adjacents:

They do not find an application and a market for the product, and end up not pursing it. And tech challenges are faced.

TESLA, NVIDIA

DISRUPTORS WITH UNREALIZED POTENTIAL

Insurgents:

They are able to build great products which are technologically advanced, and integrated systems but no real adoption

MWbotics, Agility Robotics, APPTRONIK

Incumbents:

They leverage their existing experience into building advanced robots and are successful but fail to realize adoption

ABB, KUKA, FANUC Robotics

Adjacents:

They build infrastructure to build next generation of products with minimal adoption by their user group

TESLA, NVIDIA

BUILDING A PATH TO MARKET LEADERSHIP IN HUMANOID ROBOTICS BY **LEVERAGING B2B OPPORTUNITIES, DEEP TECH AND STRATEGIC PARTNERSHIPS** BEFORE LOOKING FORWARD TO DOMINATION IN THE B2C SPACE

Drivers

AI & Hardware Integration Breakthrough

- Which player will win the race?

Public Skepticism & Ethical Concerns

- How will robots affect daily life and who will be against it?

Opportunities

- Lead AI-Hardware integration through partnerships and leveraging current expertise.
- Position Boston Dynamics as the leader of Ethical Robotics.
- Focus on customer needs where other companies continue to impress.
- Capitalize on Early Adopter markets, where technology is highly specialized and are high-trust environments.

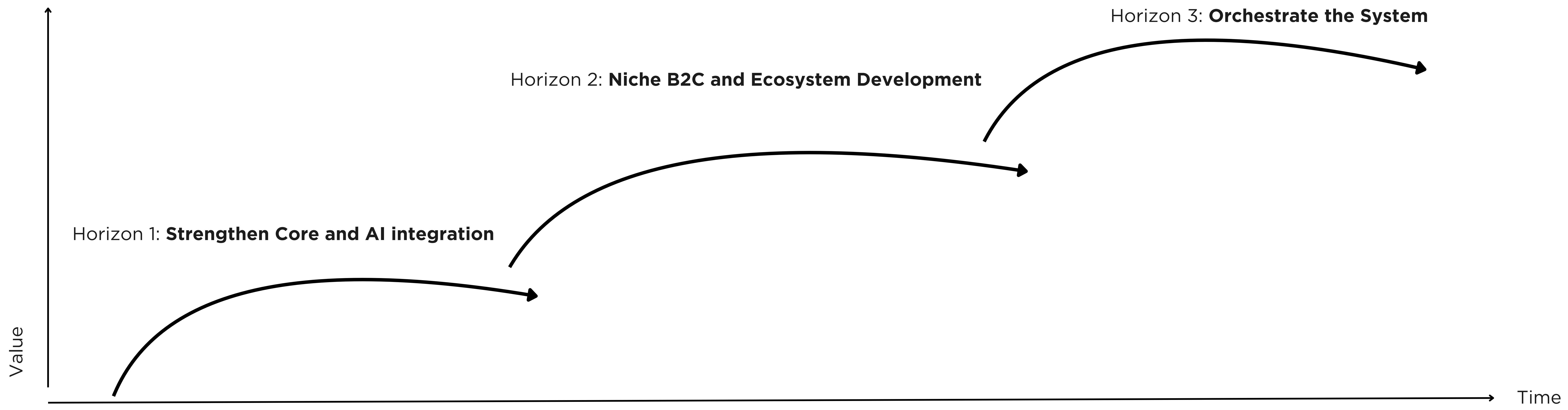
Threats

- Losing the tech race to a tech giant like Tesla/Google/Meta or even companies like Apptronik.
- Price competition from industry giants.
- Public Backlash and Public Regulations stalling growth, while all your efforts are focused on B2C.
- Over-focus on Humanoid efforts when customer needs and the market moves elsewhere.
- Fall in a “Research Trap” due to historical obsession of form over function, and not adopt a commercial mindset.

Strategy

- **Strengthen Core and AI integration:** Transition from Hardware-centric to AI-First Robotics. Focus on the ecosystem.
- **Develop Ecosystem around B2B space first:** Be a pioneer in user-centered robotics, and non-intrusive automation.
- **Orchestrate the System:** Develop a Robotics open ecosystem, later transitioning in becoming the platform for robotics.

FROM **INDUSTRIAL DOMINANCE TO CONSUMER INTEGRATION**, BOSTON DYNAMICS ENVISIONS A UNIFIED ROBOTIC FUTURE



Expand B2B solutions
- Leverage Hyundai expertise to continue cutting costs
- Launch RaaS and explore new sectors (agriculture, last-mile logistics)

Invest in AI Software Integration
- Form partnerships with leaders in AI like Nvidia to co-develop AI robotics.

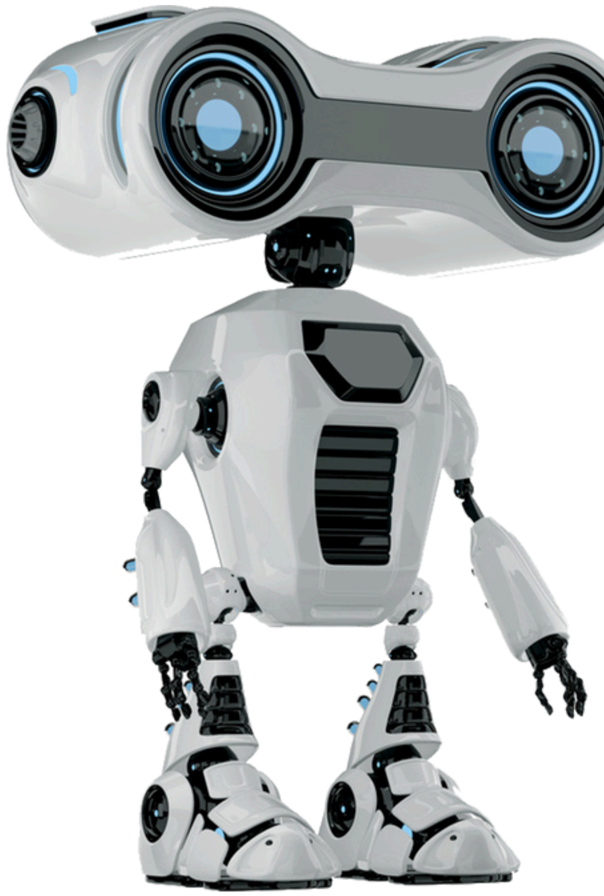
Strategic Capital Allocation
- Shift from R&D excellence to customer-driven development
- Focus on the Robotics UX team, building an ethical robotics foundation

Explore Niche B2C Markets
- Set up robust commercialization funnels to launch B2C products.
- Focus on premium service robotics (hospitality, luxury home elder care, healthcare)
- Ultra focus on JTBD, filling gaps where mass production is simply not able to cater

Pioneer a Robotics Ecosystem
- Develop an AI-powered robotics OS that integrates with smart home & IoT devices.

Transition from Hardware to Services
- License robotics intelligence and automation software
- Expand OS to outside humanoids robotics
- Become a leader in robotics and human integration, robot consulting.
- Evaluate if strategic acquisitions are necessary

Pioneer Industry Standards
- Focus on universal robotics and universal API
- Focus on all interoperability



Thank You

Appendix

Executive Summary

1. PROSPERITY IN ROBOTICS HINGES ON BRINGING **'FEATURES PROVIDABLE'** **AND** **'FEATURES DEMANDED'** TO THE SAME PAGE; NO MORE, NO LESS.
2. BOSTON DYNAMICS NEEDS TO **HARBOR AND NOURISH PUBLIC PERCEPTION** AS THEY ASPIRE TO MAKE THEIR TECHNOLOGICAL LEAPS.
3. IN AN AWAITING MARKET WITH THE CAPABILITY TO DELIVER, **A B2C MODEL** EMERGES AS A VIABLE ET LUCRATIVE OPTION.
4. A ROBOTICS COMPETITOR'S SUCCESS WILL DEPEND ON QUICKLY ALIGNING **TECH CAPABILITIES, PEOPLE'S NEED AND PRICEPOINTS**, ALL IN ONE FRAME.
5. BUILDING A PATH TO MARKET LEADERSHIP IN HUMANOID ROBOTICS BY **LEVERAGING B2B OPPORTUNITIES, DEEP TECH AND STRATEGIC PARTNERSHIPS** BEFORE LOOKING FORWARD TO DOMINATION IN THE B2C SPACE.
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